

The Anthropomorphic Trap: A Systematic Analysis of Category Errors in LLM Research

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Abstract

In spirit photographs, people perceive faces of the deceased in random photographic noise — a perceptual phenomenon known as pareidolia. This paper argues that a structurally identical mechanism operates in AI research: researchers perceive minds, personalities, and intentions in the statistical outputs of large language models (LLMs). What Floridi (2025) terms "semantic pareidolia" arises from the brain's predictive processing applying its most accessible generative model — "intentional agent" — to fluent linguistic stimuli. With anthropomorphic terminology now appearing in 40% of LLM research papers (Ibrahim & Cheng, 2025), we provide the first systematic analysis of these attributions as a unified phenomenon. We propose a taxonomy of five types of anthropomorphic category error and articulate three cross-cutting principles of invalidity — the Valid Range Principle, the Ontological Stability Principle, and the Purpose Integrity Principle — demonstrating that anthropomorphic attributions fail because the instruments are miscalibrated, the subject lacks ontological stability, and the purpose of measurement is distorted. We trace the structural causes of persistence and identify consequences including safety research distortion, policy misframing, and concrete harms to vulnerable human populations through Normative Drift and clinical tool authority erosion. Finally, we survey mathematically grounded alternative frameworks — dynamical systems theory, information-theoretic compression, topological data analysis, and reliability engineering — that describe LLM behavior with greater precision and no dependence on psychological vocabulary. The mind we see in the model remains a statistical pattern.

Keywords: anthropomorphism, category error, large language models, pareidolia, predictive processing, psychometrics, philosophy of mind, AI safety, construct validity

PART I: WHY WE FALL INTO THE TRAP

1. Introduction

1.1 The Problem: Anthropomorphism as Systematic Error

A recent empirical analysis of over 250,000 research abstracts found that anthropomorphic terminology appears in 34% of papers on large language models (LLMs), a figure that rose to 40% by December 2024 and continues to climb (Ibrahim & Cheng, 2025). Researchers routinely describe LLMs as systems that "know," "believe," "understand," "reason," and even "deceive." Papers claim these systems possess "theory of mind" (Kosinski, 2024), exhibit "personality traits" (Serapio-García et al., 2023), harbor "moral foundations" (Abdulhai et al., 2024), and organize their internal computations through "societies" of cognitively distinct

"personas" (Kim et al., 2026).

These are not merely imprecise uses of language. Each such attribution commits a specific category error — the application of concepts from one ontological domain (human psychology) to entities belonging to a fundamentally different domain (statistical text generation systems). The errors are not random; they follow identifiable patterns, arise from identifiable causes, and produce identifiable harms. Yet the field lacks a systematic framework for classifying these errors, understanding why they occur, and articulating precisely what makes them invalid.

Scattered critiques exist. Shanahan (2024) warned that mentalistic vocabulary "actively encourages anthropomorphism and obfuscates how these systems actually operate." Bender and Koller (2020) demonstrated that systems trained only on linguistic form have no principled access to meaning. Bender et al. (2021) introduced the "stochastic parrot" metaphor to counter inflated capability claims. Mitchell and Krakauer (2023) documented the heated debate over whether LLMs can "understand" language. Ullman (2023) showed that LLM performance on theory of mind tasks collapses under trivial perturbations. Gupta et al. (2024) demonstrated that personality test results for LLMs are unstable across semantically equivalent prompts. These critiques, however, remain fragmentary — each targets a specific manifestation without connecting it to the broader pattern.

1.2 Contributions of This Paper

This paper provides the first systematic analysis of anthropomorphic category errors in LLM research as a unified phenomenon. Our contributions are fourfold.

First, we propose a **taxonomy of five types** of anthropomorphic category error: Function-to-Subject Error, Construct Transfer Error, Metric Transfer Error, Intentional Vocabulary Drift, and Metaphor Reification. Each type has a distinct mechanism by which human psychological concepts are illegitimately transferred to statistical systems.

Second, we articulate **three cross-cutting principles of invalidity** that apply across all five types. The Valid Range Principle establishes that instruments calibrated for one ontological class lose construct validity when applied outside that class. The Ontological Stability Principle demonstrates that trait attribution presupposes a temporally stable subject — a condition that cloud-based LLMs structurally cannot satisfy. The Purpose Integrity Principle shows that repurposing instruments designed for welfare allocation into competitive benchmarks harms the populations those instruments were designed to serve. These principles are derived from and formalized through the case of IQ attribution to AI, but we demonstrate their generalization to all five error types.

Third, we provide a **structural analysis** of why anthropomorphic framing persists despite its invalidity, identifying five reinforcing mechanisms: citation incentives, LLM-as-judge self-reference loops, verification cost asymmetry, commercial pressures, and metaphorical inertia.

Fourth, we identify **consequences** that extend beyond research quality to include concrete harms to human populations — particularly the "Normative Drift" by which clinical instruments designed for welfare allocation are transformed into competitive benchmarks, rendering vulnerable populations statistically invisible.

1.3 Scope and Position

This paper is situated at the intersection of philosophy of science, philosophy of mind, psychometrics, and AI research methodology. It is not a contribution to the question of whether AI systems could, in principle, possess mental states — a question that remains legitimately open (Butlin et al., 2023; Chalmers, 2023). Rather, it addresses the prior question: given the current state of these systems, what category errors do researchers commit when they describe LLMs using the vocabulary of human psychology, and what are the consequences of these errors?

Our analysis is situated within the growing body of work on the logical and empirical limitations of LLMs (Bender & Koller, 2020; Mitchell & Krakauer, 2023; Shanahan, 2024), providing the first unified taxonomic framework for the category errors that this literature has identified piecemeal.

The paper proceeds as follows. Section 2 examines the cognitive mechanisms that make anthropomorphization cognitively natural and nearly irresistible. Section 3 presents the five-type taxonomy. Section 4 develops the three cross-cutting principles of invalidity. Section 5 analyzes the structural causes of persistence. Section 6 identifies consequences, including harms to human populations. Section 7 proposes alternative descriptive frameworks. Section 8 discusses implications and future directions.

2. Cognitive Mechanisms: Anthropomorphism as Semantic Pareidolia

Understanding why anthropomorphic category errors pervade AI research requires identifying the computational mechanism that generates them. We argue that the mechanism is the same one responsible for pareidolia — the perception of faces in clouds, the hearing of voices in white noise, and the seeing of the deceased in spirit photographs. The brain is a prediction machine; when its predictions misfire on a new class of stimuli, the result is systematic misattribution.

2.1 Predictive Processing and the Agent Model

Under the predictive processing framework (Clark, 2013; Hohwy, 2013; Friston, 2010), the brain is fundamentally a hierarchical Bayesian inference engine that continuously generates top-down predictions about incoming sensory signals and updates its models based on prediction errors — the discrepancy between predicted and actual input. Perception is not passive reception but active hypothesis testing: the brain asks "what in the world most likely generated this sensory pattern?" and selects the generative model that minimizes prediction error.

Among the generative models available to the human brain, the *intentional agent model* — the model that explains observed behavior by attributing beliefs, desires, and goals to its source — holds a privileged position. Koster-Hale and Saxe (2013) demonstrate that brain regions associated with theory of mind (temporoparietal junction, medial prefrontal cortex) operate through the same predictive coding architecture as perceptual regions: "predictor neurons" formulate expectations about others' behavior while "error neurons" respond to deviations. Frith and Frith (2006) identify mentalizing as a multi-component process in which the brain's mirror system, perspective-taking mechanisms, and representational capacities work together to read others' mental states — a process that, as later work in predictive social cognition has shown, operates through forward modeling analogous to motor control, where generative Bayesian models enable prediction of others' actions through inference about hidden mental states (Koster-Hale & Saxe, 2013).

The intentional agent model is not merely one model among many; it is the most cognitively accessible model for explaining complex, coherent behavior. Epley, Waytz, and Cacioppo (2007) formalize this: the self-as-agent is the default template for understanding behavior, activated automatically before alternative models are considered. When this model happens to minimize prediction error for a given stimulus — when attributing beliefs and intentions to the source of observed behavior produces better predictions than alternative explanations — the brain attributes agency, regardless of whether the source actually possesses it.

2.2 From Visual Pareidolia to Semantic Pareidolia

The phenomenon of pareidolia — perceiving meaningful patterns, especially faces and agents, in random or ambiguous stimuli — is the paradigmatic case of predictive processing misfire. In visual pareidolia, the face-detection model has such strong priors (shaped by the evolutionary importance of face recognition) that it fires even for stimuli that are not faces: clouds, toast, electrical outlets. The model minimizes prediction error locally (the pattern is somewhat face-like) while being globally incorrect (there is no face).

Floridi (2025) introduces the concept of **semantic pareidolia** to describe a structurally identical process operating in the domain of meaning rather than vision: "Traditional pareidolia is the psychological mechanism

that makes us see faces on the moon or animals in the clouds... Semantic pareidolia operates similarly, but within the realm of meaning and consciousness: we perceive intentionality where there is only statistics, meaning where there is only correlation, and understanding where there is only pattern matching on a massive scale." When AI output becomes sufficiently fluent, the brain "stops making distinctions and attributes to the artefact the properties it presents as believable."

This theoretical bridge is critical. The simulacra phenomenon in spirit photography — where bereaved individuals perceive the faces of deceased relatives in random photographic artifacts — and the anthropomorphism phenomenon in AI research — where trained scientists perceive minds, personalities, and intentions in statistical token distributions — are not merely analogous. They are manifestations of a structurally analogous computational principle — predictive processing applying its strongest prior models to ambiguous stimuli — operating at different hierarchical levels of the cognitive architecture.

2.3 Agency Detection as Evolved Bayesian Prior

Andersen (2019) provides the crucial theoretical integration by reframing Barrett's (2000) Hyperactive Agency Detection Device (HADD) in predictive processing terms. Rather than positing a dedicated agency-detection module, Andersen argues that "most false positives in agency detection can be seen as the result of top-down interference in a Bayesian system generating high prior probabilities in the face of unreliable stimuli." The HADD is not a module but a *prior* — an evolved bias in the brain's generative model that assigns high prior probability to agent-based explanations of ambiguous stimuli.

The evolutionary rationale is familiar: the asymmetric cost structure of false positives versus false negatives in predator detection selected for strong agent priors (Guthrie, 1993; Barrett, 2000, 2004). But the predictive processing reformulation adds mechanistic precision: when sensory precision is low (ambiguous stimuli), the brain relies more heavily on priors; when agent priors are strong, the result is systematic over-attribution of agency. Nees and Phillips (2015) demonstrate this empirically for auditory stimuli: contextual priming significantly increases perception of agency in ambiguous sounds, and individuals with stronger paranormal beliefs (stronger agent priors) show greater susceptibility.

2.4 Language as the Ultimate Agentive Stimulus

Natural language constitutes the most potent trigger for the agent model because language *is*, in evolutionary terms, a reliable indicator of agency. Across the entire history of our species, only minded entities have produced coherent linguistic output. The brain's generative model encodes this regularity as a near-certain prior: *if coherent language, then intentional agent*. As Reinecke et al. (2025) observe, "communication makes sense only within an agentive context" — language evolved for sharing "knowledge, thoughts, and feelings," and the brain's comprehension machinery is inseparable from the modeling of communicative intent (Grice, 1975).

LLMs violate this prior fundamentally. They produce linguistic outputs — coherent, contextually appropriate, argumentatively structured — without the mental states that language has always indexed. The prediction error generated by encountering fluent text is resolved, automatically and pre-reflectively, by applying the agent model. Shanahan (2024) captures the resulting vulnerability: "The more adept LLMs become at mimicking human language, the more vulnerable we become to anthropomorphism."

This creates what we term the **Semantic Pareidolia Problem**: the brain's predictive architecture, encountering an unprecedented category of stimulus (agent-like linguistic output from a non-agent source), applies its strongest prior (the intentional agent model) and produces systematically incorrect attributions — attributions that are computationally rational given the prior structure but ontologically mistaken. The problem is not carelessness or naivety; it is the predictable consequence of a cognitive architecture that evolved in a world where language without mind did not exist.

The Nass and Moon (2000) CASA (Computers Are Social Actors) findings and Dennett's (1987) intentional stance slippage are downstream consequences of this deeper mechanism. Humans apply social norms to

computers (CASA) because the agent model fires automatically upon encountering social cues; the intentional stance slides from instrumental to ontological because the prediction error is genuinely minimized by treating the system as an agent. The category errors documented in this paper are, in this precise sense, pareidolia — the perception of mind where there is only pattern.

PART II: WHAT THE TRAP LOOKS LIKE

3. A Taxonomy of Anthropomorphic Category Errors in LLM Research

We identify five distinct types of category error, each with a characteristic mechanism by which human psychological concepts are transferred to statistical text generation systems. While these types are analytically distinct, they frequently co-occur and reinforce one another in practice.

3.1 Type 1: Function-to-Subject Error

Definition. *A Function-to-Subject Error occurs when a functional pattern within a computational system is reified as a subjective agent possessing psychological properties.*

Mechanism. *Statistical regularities in model activations or outputs are first identified through analytical techniques (e.g., clustering, dimensionality reduction). These patterns are then assigned labels drawn from the vocabulary of human psychology — "personas," "perspectives," "cognitive styles" — effecting a two-level ontological leap: from statistical clustering to agent, and from agent to psychologically characterized agent.*

Paradigm case. Kim et al. (2026) applied sparse autoencoders (SAEs) to the internal activations of reasoning models, identified clusters of activation patterns, and labeled these clusters "personas" exhibiting "distinct personality traits and domain expertise." The paper frames the model's reasoning process as a "society of thought" involving "multi-agent-like interactions" and "debate among internal cognitive perspectives." The activation patterns are real; their interpretation as subjective agents engaging in debate is the category error. By the same logic, one could identify clusters in the activation patterns of a weather simulation and describe them as "atmospheric personalities" in "meteorological dialogue."

Literature map. The error is ubiquitous in interpretability research that moves from describing what features *do* to claiming what models *know* or *believe*. Anthropic's "Towards Monosemanticity" (2023) identifies "monosemantic features" in neural networks; while the technical contribution is descriptive, subsequent discourse frequently slides from "this feature activates in contexts related to deception" to "the model has a concept of deception." Sharkey et al. (2025) acknowledge "broader limitations of SAEs" without directly addressing the ontological inflation of their outputs.

3.2 Type 2: Construct Transfer Error

Definition. *A Construct Transfer Error occurs when a psychological construct — developed, validated, and normed for human subjects — is applied to an LLM with the assumption that the construct retains its meaning across this ontological boundary.*

Mechanism. *Human psychological instruments (tests, questionnaires, tasks) are administered to LLMs. The LLM produces responses that can be scored. The resulting scores are then interpreted as evidence that the LLM possesses the underlying psychological construct the instrument was designed to measure. The error lies in the final step: the assumption that scorability entails construct validity — that the ability to generate responses matching a scoring rubric entails possession of the psychological property the rubric was designed to detect. Lin (2025) identifies six distinct fallacies in substituting LLMs for human participants, several of which reduce to this construct transfer mechanism.*

This error manifests across three principal domains.

Theory of Mind. Kosinski (2024) administered false-belief tasks to LLMs and reported that GPT-4 solved 75% of them, "matching the performance of 6-year-old children," concluding that theory of mind "may have spontaneously emerged." Ullman (2023) demonstrated the fragility of this claim: trivial alterations — making a bag transparent, changing which agent moved an object — caused catastrophic performance drops, with GPT-3.5 scoring only 6% on modified tasks preserving the ToM structure. Sap et al. (2022) had already established that LLM social intelligence is limited: models fail systematically on tasks requiring genuine tracking of mental states. Shapira et al. (2024) invoked the "Clever Hans" analogy: LLMs may exploit spurious correlations rather than genuinely tracking mental states. Riemer et al. (2025) formalized the problem by distinguishing *literal* ToM (answering scripted questions) from *functional* ToM (adapting to agents in context), showing that LLMs pass the former while failing the latter.

Personality. Serapio-García et al. (2023) applied Big Five inventories to 18 LLMs, claiming "synthetic personality traits embedded in these models." Lee et al. (2025) concluded that "LLMs exhibit distinct and consistent personality." Three lines of critique expose the construct transfer. Gupta et al. (2024) demonstrated that semantically equivalent prompts yield statistically different personality scores. Dorner et al. (2023) found that "reverse-coded items are often both answered affirmatively by LLMs" — a pattern impossible for any entity possessing the trait. Salecha et al. (2024) showed personality scores shift by up to 1.20 human standard deviations when the model detects evaluation. Self-report instruments presuppose a self that reports — entities with subjective experience, introspective access, and temporal continuity. LLMs satisfy none of these preconditions.

Moral Foundations. Abdulhai et al. (2024) applied the Moral Foundations Questionnaire to LLMs, claiming they "exhibit particular moral foundations." Fraser (2023) documented "moral mimicry" — LLMs reproducing moral patterns from training data. Moral foundations theory explains moral intuitions as products of evolutionary pressures on embodied social beings; applying it to systems with no evolutionary history, no embodiment, and no social experience is a construct transfer.

3.3 Type 3: Metric Transfer Error

Definition. A Metric Transfer Error occurs when a measurement instrument calibrated for a specific population is applied to an entity outside that population's ontological class, and the resulting score is interpreted as a valid measurement.

Mechanism. A quantitative instrument designed for human populations (e.g., IQ tests) is administered to an LLM. A numerical score is obtained and reported as a measurement of the property the instrument was designed to measure, ignoring the fact that the instrument's validity is bound to the population for which it was designed.

Paradigm case. The widespread practice of reporting IQ scores for AI systems (OpenAI, 2023; Bubeck et al., 2023) constitutes the most pervasive instance. Ishibashi (2025a) demonstrated, using Item Response Theory, that the Test Information Function $I(\theta)$ for standard IQ tests approaches zero at the ability levels claimed for AI systems ($\theta \geq +4$), making any reported score an artifact of noise. The concept of *Structural Contamination* — where internet-available material structurally isomorphic to test items transforms fluid intelligence measures into crystallized pattern retrieval — further undermines construct validity.

Relationship to Type 2. While Type 2 errors involve transferring psychological *constructs*, Type 3 involves transferring *calibrated measurement instruments*. Type 3 carries an additional burden: numerical outputs create an illusion of precision, lending false authority to the category error.

3.4 Type 4: Intentional Vocabulary Drift

Definition. Intentional Vocabulary Drift occurs when terms from folk psychology — "knows," "believes," "understands," "intends," "deceives" — are applied to LLMs in a manner that slides from instrumentally useful shorthand to literal attribution of mental states.

Mechanism. *The drift follows a three-stage pattern. First, a researcher uses an intentional term as convenient shorthand. Second, the term is repeated, losing its "as if" qualification. Third, the literalized term enters theoretical claims and research designs.*

Shanahan (2024) distinguishes between "knowing that the word 'Burundi' is likely to succeed the words 'The country to the south of Rwanda is'" and "knowing that Burundi is to the south of Rwanda." The former describes a statistical property; the latter attributes a propositional attitude. Confusing the two is "a profound category mistake."

The "deception" literature exemplifies advanced drift. Papers on "alignment faking" (Anthropic, 2024) frame model behaviors using vocabulary presupposing intentionality and strategic goals. Koorndijk (2025) critiques this directly: "Anthropomorphism is written deeply into the concept of alignment faking as framed in the current literature."

3.5 Type 5: Metaphor Reification

Definition. *Metaphor Reification occurs when a scientific metaphor "dies" (Lakoff & Johnson, 1980) and is treated as a literal description.*

Mechanism. *Soto and Sonnenschein (2020) describe the process: "A living metaphor is simultaneously saying 'is' and 'is not.' When metaphors have been used too often, they die... reified metaphors are dangerously converted into illicit 'observables.'" Terms that were originally metaphorical — "neural network," "learning," "memory," "attention," "hallucination" — have undergone reification and now function as literal technical terms.*

The term "hallucination" is particularly revealing. Originally a metaphorical borrowing from psychiatry — denoting a perceptual experience without external stimulus, embedded in a framework of subjective experience — it smuggles in assumptions about perception and belief that are entirely unwarranted for statistical systems. Ibrahim and Cheng (2025) identify five "anthropomorphic assumptions constraining LLM development" rooted in such reified metaphors, including the assumption that LLMs "must use natural language for reasoning" and that "human values framework" is appropriate for alignment.

Relationship to Type 4. While Type 4 involves the drift of *specific terms*, Type 5 involves the deeper process by which *entire conceptual frameworks* borrowed from human psychology become embedded in the field's theoretical vocabulary.

PART III: WHY THE TRAP IS A TRAP

4. Three Principles of Invalidity

The five types of category error share a common logical structure. In this section, we articulate three principles that explain, at a general level, *why* anthropomorphic attributions to LLMs are invalid. Each principle is illustrated through the case of IQ attribution to AI (Ishibashi, 2025a; 2025b; 2025c), where the invalidity is most formally demonstrable, and then shown to generalize across all five error types.

4.1 The Valid Range Principle

Formal statement. An instrument designed and calibrated for population A loses construct validity when applied to an entity outside A's ontological class. The construct that the instrument purports to measure may not exist in the target entity; even if an analogous property exists, the instrument's calibration provides no warrant for interpreting scores as measurements of that property.

Illustration: IQ and the Information Desert. The case of IQ testing makes this principle formally precise. Ishibashi (2025a) showed, using Item Response Theory, that the Test Information Function $I(\theta) = \sum a^2 P_{\theta}(1 - P_{\theta})$ quantifies measurement precision at ability level θ . Because standard IQ test difficulty

parameters are distributed within $[-2, +2]$, the information function approaches zero at the ability levels claimed for AI ($\theta \geq +4$). Any reported score in this range is an artifact of noise. Furthermore, *Structural Contamination* — internet material structurally isomorphic to test items — transforms fluid intelligence measures into crystallized pattern retrieval, undermining construct validity entirely.

Generalization to ToM tests. False-belief tasks were calibrated for a population with a specific cognitive architecture: embodied, socially embedded, developing along a characteristic trajectory. The construct — second-order mental state representation — presupposes cognitive architecture that LLMs do not possess. Riemer et al.'s (2025) distinction between literal and functional ToM demonstrates that test scores are in the instrument's valid range numerically but outside its valid range ontologically.

Generalization to personality instruments. Self-report instruments are calibrated for entities with subjective experience and introspective access. Dorner et al.'s (2023) finding that LLMs affirm contradictory self-descriptions is evidence that the instrument operates outside its valid range — applied to an entity for which the target construct has no referent.

General theorem. For any measurement instrument M calibrated for population A , application to entity $E \notin A$ yields interpretable measurements only if: (i) the construct exists in E , (ii) M 's operationalization captures that construct with the same structural relationship as in A , and (iii) E falls within M 's calibration range. Failure of any condition renders scores uninterpretable.

4.2 The Ontological Stability Principle

Formal statement. Trait attribution presupposes a temporally stable subject whose identity persists across measurement occasions. For entities lacking this persistence, trait attribution is not merely unreliable but ontologically incoherent.

Illustration: The Heraclitean Machine. Ishibashi (2025b) illustrated this principle through the metaphor of Heraclitus's river. Classical Test Theory decomposes observed score X into True Score T and Error E ($X = T + E$), presupposing that T represents a persistent trait. Cloud-based LLMs violate this through four instabilities: (1) Update Instability — continuous silent patching creates a Ship of Theseus problem; (2) Stochastic Instability — probabilistic generation means each response is a different "collapse" of a superposition; (3) Contextual Instability — prompts mathematically restrict the accessible latent space, making the "subject" always [Model + Context]; (4) Infrastructural Instability — multi-tenant resource allocation means the "brain" is a public utility, not a private organ.

Generalization to persona attribution. The "personas" extracted by Kim et al. (2026) inherit all four instabilities. Model updates alter activation patterns; different temperatures produce different distributions; different prompts activate different latent regions. "Personas" are transient configurations of a flux system, not stable agents.

Generalization to ToM and personality. Gupta et al.'s (2024) prompt-sensitivity finding and Ullman's (2023) adversarial fragility are empirical manifestations: the "ability" being measured is not a stable trait but a prompt-dependent configuration. If the ability vanishes when surface features change, it is not a trait of the system but a property of the system-input interaction.

General theorem. For trait T to be attributable to entity E , E must possess ontological continuity — identity persistence across time. Cloud-based LLMs, as flux systems subject to Update, Stochastic, Contextual, and Infrastructural instability, fail this condition. Attribution of any trait to such systems is an ontological category error. One does not calculate the "IQ" of a random number generator; one describes its statistical distribution.

4.3 The Purpose Integrity Principle

Formal statement. Measurement instruments exist within institutional contexts and serve social purposes. Repurposing instruments in ways that ignore their institutional function distorts the measurement act itself and produces concrete harms to the populations the instruments were designed to serve.

Illustration: Doorway Diagnostics vs. Glory Badges. Ishibashi (2025c) illustrated this principle by tracing IQ testing to Binet and Simon's (1905) commission: identifying children who needed educational support. The test was a *triage mechanism* — a "Doorway Diagnostic" for allocating resources to vulnerable populations, not a "Glory Badge" for ranking superiority. When AI laboratories report IQ scores, they invert this purpose, producing **Normative Drift**: the median (IQ 100) is reframed as a "baseline for competence," casting the 50% of humanity below the median as "sub-optimal."

Generalization to ToM testing. ToM tests were developed to identify children with autism spectrum conditions. Celebrating LLMs for "passing" these tests at "the level of a 6-year-old" transforms a diagnostic tool for neurodevelopmental conditions into a machine capability benchmark, risking trivialization of the lived experience of individuals for whom these tasks represent genuine challenges.

Generalization to personality testing. Assigning "psychopath" scores to LLMs (Li et al., 2022) repurposes clinical instruments as sensationalist labels, trivializing personality disorders and reinforcing stigma.

Generalization to intentional vocabulary. "Deception" carries institutional weight in law, ethics, and interpersonal relations — it assigns moral responsibility to intentional agents. Applying it to statistical systems obscures the actual locus of responsibility: developers, deployers, and infrastructure operators.

General theorem. For any instrument M with institutional purpose P serving population Q, repurposing M for purpose P' ≠ P applied to entity E ∉ Q: (i) distorts the measurement act by stripping interpretive context, and (ii) risks harming Q by subordinating M's welfare function to P's objectives.

4.4 The Three Principles as a Unified Framework

The three principles are logically independent but empirically reinforcing. The Valid Range Principle establishes that the *instrument* fails. The Ontological Stability Principle establishes that the *subject* fails. The Purpose Integrity Principle establishes that the *purpose* fails. Together, they constitute a complete critique: anthropomorphic measurement of AI is invalid because the tool is miscalibrated, the object is ontologically incoherent as a subject of measurement, and the act of measurement is a misappropriation of social infrastructure.

Principle	What Fails	Logical Level	Key Question	-----	-----	-----	-----	Valid
Range	Instrument	Epistemological	Does the tool measure what it claims?			Ontological Stability	Subject	
Ontological	Is there a stable subject to measure?		Purpose Integrity	Purpose	Ethical/Institutional	Should	this tool be used this way?	

This framework applies not only to formal testing but to any act of anthropomorphic attribution. When a researcher writes "the model understands X," they implicitly apply an instrument (folk-psychological "understanding") outside its valid range (human cognition), to an unstable subject (a flux system), for a purpose (describing AI capabilities) that distorts the concept's original function (characterizing human cognitive states).

PART IV: WHY THE TRAP PERSISTS

5. Structural Causes of Persistence

If the anthropomorphic trap is as pervasive and as logically flawed as the preceding analysis suggests, why does it persist? The answer lies in structural features of the AI research ecosystem that systematically reward anthropomorphic framing.

5.1 The Citation Economy

Anthropomorphic claims generate attention. "Theory of Mind May Have Spontaneously Emerged in Large Language Models" attracts citations and media coverage; "Statistical Associations Between LLM Outputs and ToM Task Surface Features" does not. McGreivy and Hakim (2024) document that 79% of papers claiming machine learning superiority use "weak baselines," driven by publication incentives. Ibrahim and Cheng's (2025) data confirms the trend: anthropomorphic terminology in LLM abstracts grew from 34% to 40% in one year with no correction mechanism. Natale (2024) links this to funding: anthropomorphic discursive practices "reflect the vested interests of stakeholders."

5.2 The LLM-as-Judge Self-Reference Loop

When LLMs evaluate LLM behavior, a triple amplification occurs. Researchers' anthropomorphic intuitions shape evaluation prompts; the LLM evaluator reflects anthropomorphic patterns from training data; results are interpreted through the same anthropomorphic lens. ICLR 2025 documented systematic biases: GPT-4 favors its own outputs with 10% higher win rates. The loop is impervious to disconfirmation from outside the system. When researchers ask GPT-4 to evaluate whether another model "understands," they ask a system trained on human descriptions of understanding to judge whether outputs match those descriptions — embedding the category error in the methodology.

5.3 Verification Cost Asymmetry

Anthropomorphic claims enjoy a structural advantage: they are intuitively plausible and expensive to disprove. Reporting that an LLM "understands sarcasm" resonates with user experience; disproving it requires adversarial examples, philosophical argument, and labor-intensive controls. We observe that within AI research, this asymmetry may compound: researchers who produce anthropomorphic claims gain visibility through those claims, while the cost of rigorous disconfirmation discourages correction.

5.4 Commercial Incentives

"This AI has an IQ of 140" is a sales pitch; "This statistical system achieves 95.2% accuracy on next-token prediction benchmarks under specified conditions" is not. Deshpande et al. (2023) warn that "anthropomorphized LLMs customized for different user bases violate multiple provisions in the legislative blueprint" — yet commercial incentives to anthropomorphize persist because they drive engagement.

5.5 Metaphorical Inertia

The field suffers from *metaphorical inertia*: decades of anthropomorphic vocabulary embedded in technical infrastructure, documentation, methodology, and pedagogy. "Neural network," "attention," "learning," "memory" constitute the shared vocabulary of the research community. Replacing them would require conceptual reconstruction — replacing an entire metaphorical framework with one equally communicable but less misleading. The transition cost creates powerful status quo bias.

PART V: WHAT WE LOSE

6. Consequences of the Anthropomorphic Trap

6.1 Safety Research Distortion

When AI safety frames risks anthropomorphically, it prescribes agent-appropriate solutions for system-generated problems. If an LLM "deceives," one constrains the agent's intentions — presupposing the intentionality in question. If the system produces divergent outputs correlated with training distributions, the appropriate response is statistical monitoring and distributional analysis — a fundamentally different paradigm. Dahlgren Lindström et al. (2025) critique RLHF as potentially "good for reinforcing anthropomorphic behaviour in LLMs" without increasing actual safety. The HHH framework itself —

Helpful, Harmless, Honest — presupposes properties of *agents*, not statistical distributions.

6.2 Policy and Regulatory Misframing

Anthropomorphic descriptions flow into policy discourse. When policymakers read that AI "understands," "reasons," and "deceives," they construct regulatory frameworks for agents — assigning rights, responsibilities, and accountability structures for entities with intentions. Peter et al. (2025) argue this creates challenges for determining regulatory scope and enforcement. A framework designed to govern "deceptive AI" focuses on constraining agent capacities, missing the actual problem: training data and optimization objectives producing outputs humans interpret as deceptive.

6.3 Resource Misallocation

If LLMs are entities with minds, understanding AI becomes cognitive science. If LLMs are complex statistical systems, understanding AI becomes dynamical systems analysis and information theory. The anthropomorphic frame has directed substantial resources toward the former paradigm. Resources spent testing whether LLMs "possess theory of mind" are resources not spent characterizing transformer activation dynamics or developing robust failure prediction.

6.4 Human-Side Harm

The least examined consequence is the impact on human populations — particularly vulnerable ones.

Normative Drift. Comparing AI scores to human norms reframes the statistical median as a minimum standard. When AI "surpasses the human average," the implicit message is that the average is a baseline to be exceeded. This marginalizes the approximately 15% of the population with IQs below 85 — individuals who rely on the clinical instruments being repurposed as benchmarks — by recasting them as below a "standard" never intended to function as one.

Clinical Tool Authority Erosion. IQ tests, ToM assessments, and personality inventories serve gatekeeping functions in educational, clinical, and legal settings. When publicly associated with AI benchmarking — "GPT-4 Passes Theory of Mind Test" — their clinical authority erodes. The institutional seriousness of evaluation is undermined.

Stigma Amplification. When researchers report LLMs score high on "psychopathy" or "Dark Triad" measures (Li et al., 2022), clinical terms circulate as entertainment — casual markers of interesting model behavior rather than serious designations with real consequences for diagnosed individuals.

These harms are predictable consequences of the Purpose Integrity Principle: when instruments designed for welfare allocation are repurposed as competitive benchmarks, the populations they serve become collateral damage.

PART VI: HOW TO ESCAPE

7. Alternative Descriptive Frameworks

The preceding sections have diagnosed the problem; this section surveys emerging solutions. The alternatives are not speculative proposals but active research programs, each with growing literatures, empirical validation, and — crucially — no dependence on psychological vocabulary. They share a common logic: describe what LLMs *do* using concepts from the mathematical and engineering sciences that were developed *for* the kinds of systems LLMs actually are.

7.1 Dynamical Systems Description

The most mathematically rigorous alternative treats transformer layers not as stages of "thought" but as time-steps in a dynamical system. Geshkovski, Letrouit, Polyanskiy, and Rigollet (2025) develop this view in a landmark paper published in the *Bulletin of the American Mathematical Society*, proving that self-attention dynamics are equivalent to interacting particle systems on the unit sphere. Tokens are particles; layers are time; and the long-time behavior is *clustering* — self-attention matrices converge to low-rank Boolean matrices. This mathematical framework replaces "the model synthesizes information" with "the particle system converges to a clustered equilibrium under mean-field dynamics." The descriptive precision is superior; the ontological inflation is zero.

Empirical work corroborates the theoretical picture. Gurnee and Tegmark (2024) analyze residual streams as dynamical systems, identifying attractor-like dynamics and curved trajectories in reduced phase space. Fernando and Guitchoyts (2025) extend this to Llama 3.1 8B, documenting rotational dynamics, unstable periodic orbits, and attractor-like convergence — while explicitly noting that their dynamical metrics "capture formal properties and do not imply subjective experience." Wang, Li, Yan, Cheng, and Zhang (2025), in an ACL 2025 long paper, demonstrate that successive paraphrasing by LLMs converges to stable 2-period attractor cycles, a finding that reframes "limited creativity" as a measurable property of the system's basin of attraction.

The dynamical systems vocabulary also provides tools for characterizing failure. Chen et al. (2025) introduce a Quasi-Lyapunov Exponent (QLE) for transformer layers, quantifying sensitivity to initial perturbation — the "butterfly effect" in LLM reasoning. Small input perturbations propagate nonlinearly, confirming that the system exhibits chaotic dynamics. Where anthropomorphic framing describes this as "the model gets confused by slight rewording," the dynamical framing offers a quantitative metric: positive QLE indicates chaotic sensitivity in the layer-wise information accumulation process. Ugail and Howard (2026) apply detrended fluctuation analysis and metastability metrics to characterize functional regimes in GPT-2, providing a regime classification system entirely free of cognitive metaphor.

7.2 Information-Theoretic Description

Information theory provides a second framework grounded in mathematics rather than folk psychology. Delétang et al. (2024) demonstrate that language modeling is, formally, compression. Performance measured through bits-per-token efficiency is grounded in Shannon's framework; "the model understands grammar" becomes "the model's compression ratio improves when input conforms to grammatical regularities." Mathematical precision makes the ontological inflation of anthropomorphic vocabulary visible by contrast.

The Minimum Description Length (MDL) principle extends this logic from evaluation to explanation. Lan, Chemla, and Katzir (2024) show that MDL-optimized neural networks achieve provably correct solutions for formal language learning — and that these correct solutions are optima under MDL but *not* under standard training objectives. This result has direct implications for the anthropomorphic trap: what is typically described as "the model learns the rule" can be reformulated as "the MDL-optimal compression of the training data entails the formal language's structure." The explanatory gain is real; the psychological vocabulary is eliminable.

At a deeper level, Wan et al. (2025) unify scaling laws through conditional Kolmogorov complexity, proving that LLM training approximates the upper bound of joint Kolmogorov complexity of training data. They further demonstrate that infinite problems exist that *cannot* be solved by LLMs on this analysis — a principled limitation invisible to anthropomorphic framing. The compression framework thus offers not only better description but better prediction: it identifies the boundaries of LLM capability through information-theoretic impossibility results rather than vague intuitions about "lack of understanding."

7.3 Topological and Geometric Description

A third framework characterizes LLM internal representations through geometry and topology rather than semantic interpretation. Li and Sarwate (2025) propose a stratified manifold hypothesis: LLM embedding spaces are not single smooth manifolds but unions of lower-dimensional strata whose intrinsic dimensions

vary with input domain and perplexity. Using a Mixture-of-Experts framework with varying sparsity levels, they demonstrate that expert assignments cluster along semantically interpretable boundaries — described entirely in geometric terms (intrinsic dimension, gating entropy, inter-expert distance) without recourse to claims about what the model "knows."

Topological Data Analysis (TDA) provides still finer tools. Fay, García-Redondo, Wang, Dubossarsky, and Monod (2025) apply persistent homology to hidden states across six LLMs, computing 41-dimensional feature vectors from persistence barcodes that capture "topological compression" under adversarial influence. Fitz, Romero, and Schneider (2024) introduce "perforation" as a measure of topological complexity in LLM hidden layers, tracking how Betti numbers evolve across depth and training time. These tools describe the *shape* of internal representations — connected components, loops, and higher-dimensional voids — without any assumption about what those representations "mean" in human terms.

The geometric approach also provides a striking quantitative signature distinguishing human from machine text production: human-generated text has intrinsic dimension approximately 9 for alphabetic languages, while LLM-generated text shows intrinsic dimension roughly 1.5 lower (Tulchinskii et al., 2024). This is a physical measurement, not a psychological judgment — and it reveals a structural difference that anthropomorphic framing renders invisible.

7.4 Reliability Engineering and Specification Frameworks

For deployment contexts, reliability engineering provides an evaluation framework borrowed from industrial process control rather than cognitive science. Chu, Talluri, Lu, and Iosup (2025), in a paper at ACM/SPEC ICPE 2025, analyze 8 LLM services (including ChatGPT and Claude) using classical dependability metrics: Mean Time Between Failures (MTBF), Mean Time To Recovery (MTTR), and failure-recovery process stages. Their findings — median MTBF of 1.74 to 16.80 days across services — describe LLMs as industrial systems requiring the same reliability analysis as power grids or telecommunications infrastructure, not as minds requiring psychological assessment.

Ranganathan, Pathak, and Bansal (2025) extend this approach to the inference layer, analyzing 156 high-severity production incidents and developing a taxonomy of failure modes: inference engine failures (60% of incidents), timeouts (40% of engine failures), resource exhaustion, and configuration drift. Vinay (2025) identifies 15 hidden failure modes in LLM systems, including multi-step reasoning drift, context-boundary degradation, and version drift, explicitly framing "LLM reliability as a system-engineering problem rather than a purely model-centric one."

This engineering perspective yields a concrete alternative to anthropomorphic benchmarks: the specification sheet. Gebru, Morgenstern, Vecchione et al. (2021) established the paradigm with "Datasheets for Datasets," adopting the electronics industry convention of documenting operating characteristics, test conditions, and recommended uses. The approach replaces "How intelligent is this AI?" with "Under what operating conditions does this system meet its specified tolerances?" — a question that is both more precise and more useful. Chollet (2019) anticipated this shift with the Abstraction and Reasoning Corpus, proposing criterion-referenced evaluation: "Does this system solve task X with >95% accuracy across 10,000 trials under context Y?" rather than "Is this AI as smart as a 6-year-old?"

7.5 Functional Description Discipline

Even in informal contexts, a discipline of functional description can mitigate error. Shanahan (2024) proposes: describe LLMs as generating "statistically likely continuations" rather than "thinking"; use quotation marks when intentional terms are unavoidable; ask "What does the system actually do?" before using psychological vocabulary. The key distinction is between "behaves as if" and "is" — preserving the intentional stance's predictive utility while blocking ontological commitment.

7.6 Neologism and Subscript Notation

The field may need new vocabulary — neologisms acknowledging the novelty of LLM capabilities without mapping them onto human concepts. Subscript notation (e.g., "understanding_M") marks functional overlap while preserving the ontological boundary that anthropomorphic vocabulary erases. The resistance to neologism is understandable; the cost of not creating new vocabulary is the perpetuation of category errors. The history of science offers precedent: the transition from phlogiston to oxygen required not just new theories but new words.

8. Discussion and Conclusion

8.1 Conceptual Hygiene as a Field-Wide Imperative

This paper has documented a systematic pattern of category errors — not scattered imprecisions but a coherent syndrome with identifiable types (Section 3), explicable invalidity (Section 4), structural causes (Section 5), and concrete consequences (Section 6). The three principles — Valid Range, Ontological Stability, and Purpose Integrity — yield practical diagnostics. Researchers can ask of any attribution: (1) Is the concept applied within its valid range? (2) Does the entity possess the stability required for the attributed property? (3) Does the attribution preserve or distort the institutional purpose of the concept?

If the answer to any question is negative, the attribution commits a category error.

Crucially, Section 7 demonstrates that alternatives are not only available but mathematically superior. The dynamical systems, information-theoretic, topological, and reliability engineering frameworks surveyed there describe LLM behavior with greater precision, superior predictive power, and zero ontological inflation. The choice is not between anthropomorphic description and silence; it is between a vocabulary borrowed from folk psychology and vocabularies purpose-built for the mathematical and engineering properties these systems actually possess.

A pragmatist might object that anthropomorphic vocabulary, while ontologically imprecise, serves legitimate instrumental purposes: prediction, communication efficiency, and conceptual scaffolding (Peeters et al., 2021). Lederman and Mahowald (2024) sharpen the point by distinguishing "library" and "librarian" framings — arguing that the choice of metaphor shapes which research questions are asked. We do not deny the instrumental value of the intentional stance (Dennett, 1987). Our argument is that the instrumental benefits are achievable without ontological inflation — the alternative frameworks in Section 7 preserve predictive and communicative utility while eliminating the category errors that produce the harms documented in Section 6. The transition we advocate is not from description to silence but from a vocabulary that systematically generates category errors to vocabularies that do not.

8.2 Why the Trap May Be Cognitively Inescapable

The persistence of anthropomorphic error documented in Sections 3–6 raises a deeper question: is the trap merely a correctable habit, or does it reflect something more fundamental about human cognition? Section 2 showed that anthropomorphism functions as semantic pareidolia — the predictive processing architecture applies its most accessible generative model ("intentional agent") to ambiguous stimuli. But the problem may run deeper still.

Consider the parallel with causal cognition. Ishibashi (2026) argues that the sense of "causation" itself arises from prediction error minimization: the brain labels highly predictable regularities as "causal," such that causation may be not a feature of reality but a cognitive label applied to reliable covariation. If this analysis is correct, the anthropomorphic trap is a special case of causal pareidolia. When an LLM reliably produces outputs consistent with "understanding," the brain's predictive machinery labels this regularity as *caused by* understanding — not because the researcher is careless, but because the cognitive architecture is functioning exactly as designed: it conflates reliable covariation with causal explanation. The "recognition" of mind in LLM outputs is not a failure of cognition but its normal operation. We do not *fail* to see that LLMs lack minds; we *succeed* in applying the causal label "mind" to a stimulus that reliably covaries with the patterns

minds typically produce. The match is genuine at the level of covariation; the error lies in treating a cognitive label as an ontological discovery.

This analysis has a practical implication: exhortations to "stop anthropomorphizing" may be insufficient precisely because they ask researchers to override a cognitive default that conflates reliable prediction with causal explanation. The alternative frameworks surveyed in Section 7 succeed not by suppressing the anthropomorphic impulse but by *replacing the template* — providing dynamical, information-theoretic, and topological patterns that compete with the "mind" template for cognitive resources. The transition from phlogiston to oxygen did not require chemists to stop seeing combustion; it required them to see combustion *as something else*. The same logic applies here.

8.3 Limitations and Future Directions

This paper has focused on current LLMs — systems that are, by best evidence, statistical text generation engines without subjective experience or intentional states. Should future systems satisfy indicator properties derived from scientific theories of consciousness (Butlin et al., 2023), the framework would need recalibration, not abandonment.

Future work should pursue: empirical validation of the taxonomy through systematic paper coding; operationalization of the three principles into peer review checklists; empirical investigation of the human-side harms — Normative Drift, clinical tool erosion, stigma amplification — through survey and interview methods; and development of standardized specification sheets for LLM evaluation that adopt the engineering frameworks surveyed in Section 7, replacing anthropomorphic benchmarks with dynamical, information-theoretic, and topological metrics.

The anthropomorphic trap is not a minor terminological quibble. It is a systematic distortion of how a significant portion of the research field understands its object of study, with consequences reaching from methodology to policy to the lived experience of vulnerable populations. Escaping it requires recognizing that sophisticated linguistic behavior does not entail the cognitive architecture that typically produces such behavior in humans — a category distinction that the field has systematically failed to maintain, and that the three principles articulated here provide the tools to enforce.

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